

SMART IRRIGATION & PLANT-CARE SYSTEM

PROJECT VERDE

The plant that waters itself — and talks to AI.

DESIGNED & BUILT BY

Aarav Choudhary • Anuj

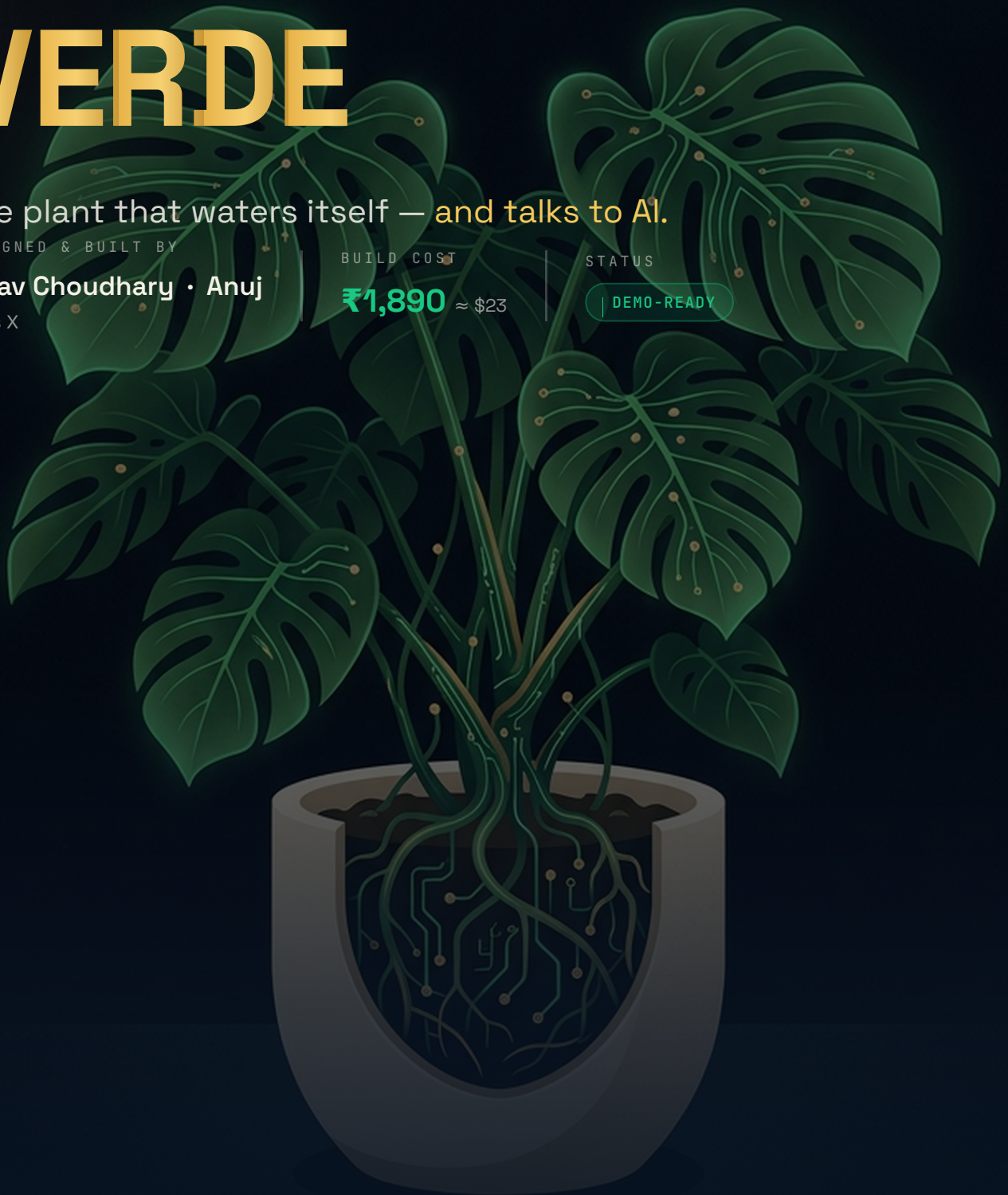
Class X

BUILD COST

₹1,890 ≈ \$23

STATUS

DEMO-READY



The whole story in 60 seconds

Everything a judge — or a parent — needs to know about Project Verde, compressed onto one page. Details wait patiently in the chapters that follow.

City plants die quietly — of forgetfulness and guesswork. **Project Verde** is a three-tier IoT system that lets a plant **say** what it needs and then acts on it: sensors on an ESP32 read the soil, air, light and water tank every second, a Firebase cloud keeps the truth, and a web app turns it into a dashboard, a weather brain and an AI plant doctor.

₹1,890

TOTAL BUILD • ≈ \$23

Full working product for a quarter of a commercial kit's price. Software & APIs: free tiers only.

5 + 2

5 SENSORS • 2 ACTUATORS

Soil, air, light, tank — watched 24/7. A pump that waters and a UV grow light that glows.

1 sec

CLOUD HEARTBEAT

Ten metrics upstream, nine commands downstream — every single second, without a hiccup.

94%

AI PLANT-DOCTOR ACCURACY

Snap a photo in-app; crop.health named the nutrient deficiency at 94% with a treatment plan.

17 → 2

CLOUD CALLS PER SECOND

The bug that nearly sank us, and the JSON-bundling fix that cut latency ~85%. Page 14.

0

REBOOTS IN 10-MIN SOAK

Hardware watchdog held the line through a 10-minute torture test. 13/13 system tests pass.



The Edge

ESP32 brain + 5 sensors, pump & UV light. ESP32-CAM eyes.



The Cloud

Firebase Realtime DB — one source of truth, synced each second.



The Experience

Single-file web app: dashboard, weather, plant doctor, AI chats.

“

The plant doesn't wait for you to remember. It tells the cloud — and the cloud tells us.

— THE ONE-SENTENCE PITCH

Contents

Thirty-three pages, eight chapter-rooms. Every line below is a live link — divider art marks each turn of the story.

01 The story in 60 seconds 02 One page. Every number that matters.	02 Why plants die in cities 04 The problem — and the ₹8,000 gap in the market.
03 Our solution, in one loop 06 Sense → decide → act → explain. Three tiers, one plant.	04 How it works 07 System architecture · the one-second heartbeat timeline.
05 Hardware 09 Meet the modules · wiring map · power lessons learned the smoky way.	06 Firmware — the engine room 12 Non-blocking scheduler · AUTO logic · watchdog discipline.
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How to read this document



In a hurry? Pages 2, 14 and 26 tell the whole story. Here for engineering? Chapters 05–08. Every page is built to be skimmed in under ten seconds — headings carry the plot.

SKIM-FRIENDLY
BY DESIGN

02

The silence that kills houseplants.

City plants die quietly — of forgetfulness and guesswork. Before a single wire was cut, we needed to feel that problem: the forgotten fern, the drowned succulent, the blind owner watering on vibes.

01a-div-why.html0

Plants don't die of thirst. They die of silence.

Nobody kills their plant on purpose. They kill it with a busy week — because a plant has no voice, no notification, no way to say "I'm dry," "you're drowning me," or "the tank is empty."



Forgotten

Urban families travel, study late, work shifts. Watering slips a day, then a week. The soil says nothing — it just quietly hits zero.



Drowned

The opposite sin: daily "just in case" watering. Roots rot in soggy soil while the plant looks, from above, perfectly loved.



Blind

No one knows the soil moisture right now, whether the tank is empty, or that rain is coming tonight. Plants die of missing information.

₹8,000+

COMMERCIAL SMART-GARDEN KITS

And for that money? **No camera.**
No AI doctor. A sealed black box a student can't open, understand, or repair. We priced the parts at our local market and did the maths: the gap between "product" and "project" wasn't money — it was information.

4.2×

CHEAPER TO BUILD VERDE

A WEEK IN THE LIFE OF A FORGOTTEN FERN

MON	TUE	THU	SAT	SUN
61%	48%	29%	14%	
Feels fine. Was watered "recently". Nobody wrote down when.	Still green. Still silent. Family leaves for a short trip.	Below the comfort line. A sensor would have texted by now.	Leaves curl. The plant is shouting — in a frequency nobody hears.	Everyone returns to a funeral. Cause of death: missing data.

“

he plant knew it was thirsty for days. We found out at the funeral.

— EVERY PLANT PARENT, EVER



03 · OUR ANSWER

THE LOOP

So we built a plant that speaks up.

Verde closes the loop a human keeps forgetting to close. Four verbs, running every second of every day — for roughly the cost of four movie tickets.



1 • Sense

Five probes read soil, air, light & tank — filtered, averaged, honest.



2 • Decide

The ESP32 brain applies AUTO logic with thresholds you tune live.



3 • Act

Pump waters until the soil says "enough" — then stops, exactly there.



4 • Explain

Dashboard, weather veto, camera, AI doctor. The plant finally has a voice.

TIER 1 • EDGE

ESP32 WROOM-32 + five sensors, relay-driven pump, UV grow LED. ESP32-CAM takes the photos. No cloud needed to keep the plant alive — AUTO runs on-device.

TIER 2 • CLOUD

Firebase Realtime Database: the single source of truth. Sensors, controls, scans, weather and logs — one JSON tree, public-read, validated writes.

TIER 3 • EXPERIENCE

A single-file web app — dashboard, weather, plant doctor, AI assistants — talking to four external APIs on free tiers.

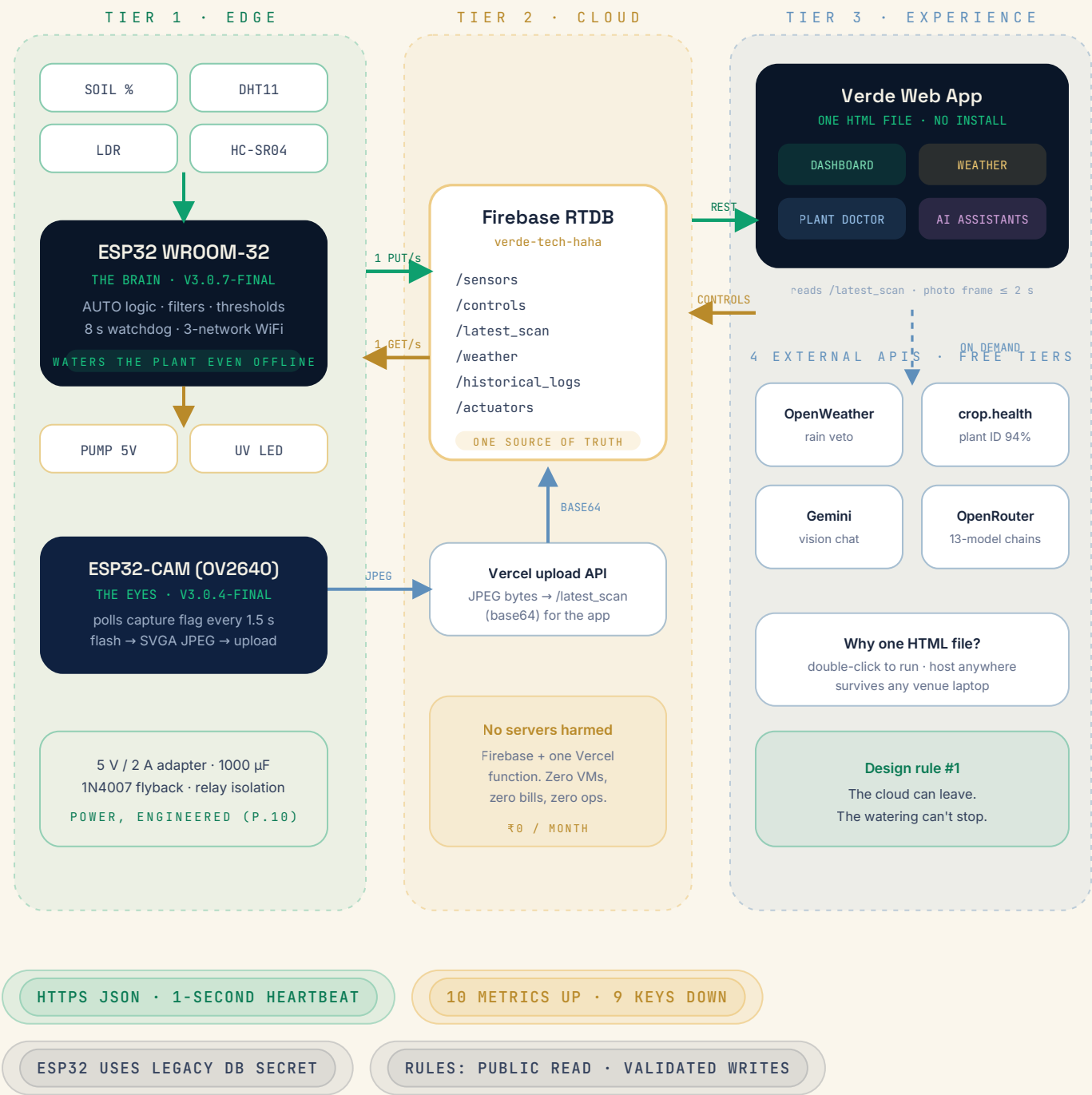
₹1,890 TOTAL ≈ \$23

WORKS OFFLINE-OF-CLOUD IN AUTO

STUDENT-OPENABLE · STUDENT-FIXABLE

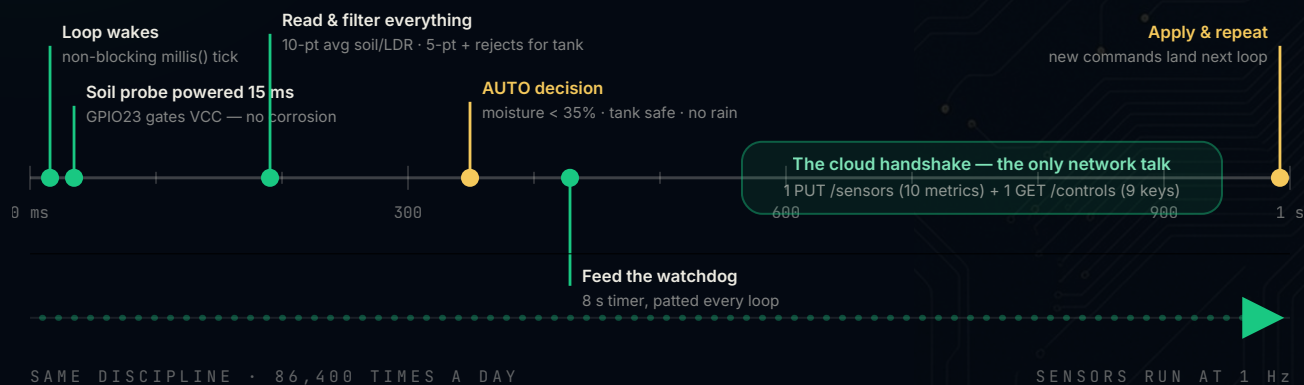
Three tiers, one conversation

Hardware at the edge, truth in the cloud, delight on the screen. If the internet dies, the plant still gets watered — AUTO logic lives on the chip, not in the cloud.



The one-second heartbeat

Every second, forever, the same disciplined routine. No `delay()`, no blocking calls, no excuses — a `millis()` scheduler keeps every task in its lane.



1 s

Sensors + cloud

Fresh metrics out, fresh commands in. The dashboard never sits on stale data.

1.5 s

CAM polls

The camera asks "photo requested?" — after the main brain, never blocking it.

10 s

WiFi health

Drop? Reconnect. Falls back across home → hotspot → school networks.

60 s

History logging

A moisture point lands in `/historical_logs` every minute — fuel for the trend chart and tomorrow's predictive watering.

3 min

Weather check

The app re-reads Delhi's sky. Rain brewing? A countdown starts, then the pump is vetoed. Page 17.

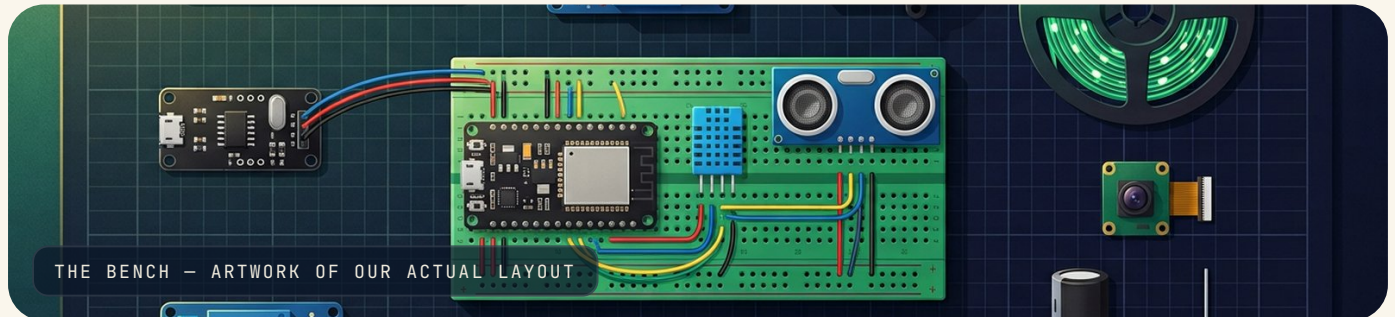


Why the obsession with "non-blocking"?

Because the one time we let network calls block the loop, the watchdog ate us — a reboot every 10 seconds and a pump that stuttered like a telegraph key. That story gets its own spread: page 15.

Five senses, two hands, two brains

Every module earned its place on the breadboard. Pins below are exactly what the firmware speaks to.



ESP32 WROOM-32 — the brain

32-bit • WiFi

Runs AUTO logic, filters, thresholds and the 1-second heartbeat. Keeps watering even if WiFi dies.



ESP32-CAM — the eyes

OV2640

Shoots SVGA plant portraits on demand, uploads within seconds. Own board, own firmware, zero load on the brain.



Soil moisture

A0 → GPIO 34

VCC gate → GPIO 23

LM393 comparator, read for just 15 ms at a time — power-gating stops probe corrosion.



DHT11

DATA → GPIO 4

Air temperature + humidity. Cheap, slow, honest — sampled once a second.



LDR module

A0 → GPIO 35

Sees dusk arrive. "Dark" triggers the grow light; $\pm 2\%$ hysteresis stops flicker at sunset.



HC-SR04

TRIG → 18

ECHO → 19

Ultrasonic tank level. 5-point filter rejects pump-splash garbage — a splash can't fake an empty tank.



2-ch relay → pump

IN1 → GPIO 5

ACTIVE-LOW

Switches the 5 V submersible pump on its own isolated supply. The brain never touches pump current.



UV grow LED

GPIO 12

ACTIVE-HIGH • 220 Ω

Photosynthetic light on schedule or on command — sunrise on demand for a windowsill plant.



Why two microcontrollers?

The camera is a diva: it wants RAM, clean power and precise timing. Give it its own board and the brain stays fast — one waters, one watches, neither waits for the other.

Every wire, accounted for

Signal wires in emerald, power in gold, ground in slate, ultrasound in blue. This is the map we wish we'd taped to the table on day one.



Power: the chapter written by smoke

Nothing in this section came from a tutorial. Each lesson cost us an afternoon, a component, or a small heart attack.

LESSON 01 • THE 67 W TRAP

A laptop charger is not a power supply

We plugged a 67 W USB-PD brick into the board and got... nothing. ~0 mA. USB-PD negotiates voltage through a handshake chip the ESP32 doesn't have, so the charger politely offered 5 V at a trickle. The fix was gloriously dumb: a **5 V / 2 A phone adapter**. Dumb power is honest power.

LESSON 02 • THE SAG ABSORBER

1000 μ F of forgiveness

A pump starting and a WiFi burst both demand current now. The rail dipped, the ESP32 browned out, the boot looped. A **1000 μ F electrolytic capacitor** strapped across 5 V/GND eats those spikes whole.

LESSON 03 • THE INDUCTIVE KICK

1N4007: the bodyguard diode

Switch a motor off and its coil fires a voltage spike backwards down the wire — straight at the GPIO. A **1N4007 flyback diode** across the pump gives that spike a safe loop to die in.

LESSON 04 • THE ISOLATION WARD

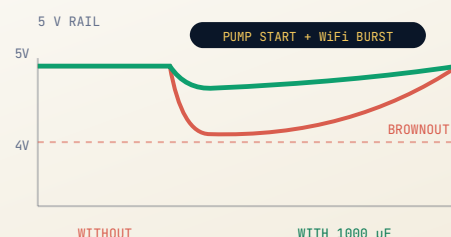
The pump lives on its own circuit

The relay's COM/NO contacts put the pump on a **separate 5 V source**. Logic side and motor side share a ground reference and nothing else — pump noise never flows through the brain.

THE MYSTERY, SOLVED

"PSRAM not found" was never about RAM

The CAM refused to boot, screaming about missing PSRAM. The RAM was fine — the **power wasn't**. Weak 5 V = brownout during camera init. The 5 V / 2 A adapter fixed a "memory error" without touching a single line of code. Hardware bugs love to dress up as software bugs.



Our power checklist (tape it to your table)

Dumb 5 V / 2 A adapter • 1000 μ F across the rails • diode across the pump • pump on relay-isolated supply • rails bridged + to +, - to -. Five lines, zero reboots.

06

Where copper traces flow like water canals.

An ESP32 sketch that behaves like a grown-up RTOS — every task on a schedule, every sensor filtered, every decision logged. This is the chapter with the smoke stains.

[04z-div-firmware.html](#)

The engine room

Code_1_Main_Brain.ino • V3.0.7-FINAL

An ESP32 sketch that behaves like a grown-up RTOS: every task on a schedule, every sensor filtered, every decision logged. No blocking calls anywhere — not after what happened on page 15.

1 Hz

SENSOR READS

All five probes sampled, filtered and judged once per second.

1 s

CLOUD SYNC

One bundled write, one bundled read. The heartbeat itself.

10 s

WIFI HEALTH

Reconnect patrol across 3 saved networks: home, hotspot, school.

60 s

HISTORY LOG

A moisture point banked every minute for the trend chart.



The watchdog pact

A hardware watchdog arms an **8-second** fuse. The loop must pet it every pass; a hung loop gets a reboot instead of a dead plant. 10-minute soak test: **0 reboots**.



Filters that don't lie

10-point moving averages for soil & LDR. The tank gets a **5-point average + invalid-read rejection** — pump splash can't impersonate an empty tank.



±2% hysteresis

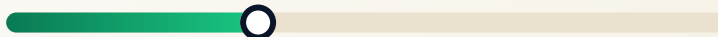
At sunset the light sensor hovers on the edge. Hysteresis gives the grow LED a dead-band so it glides on once — **no flicker disco**.



Thresholds you can tune from the app — saved to flash (NVS)

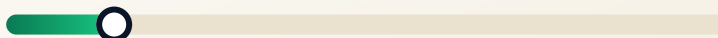
SURVIVES POWER CUTS

SOIL MOISTURE TRIGGER



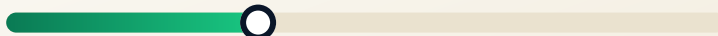
35%

TANK MINIMUM (0 = OFF)



15%

LIGHT / DARK LINE



35

Slide these in the app and the ESP32 adopts them mid-loop — no reflash, no serial monitor, no ceremony.

3-NETWORK WIFI FALLBACK

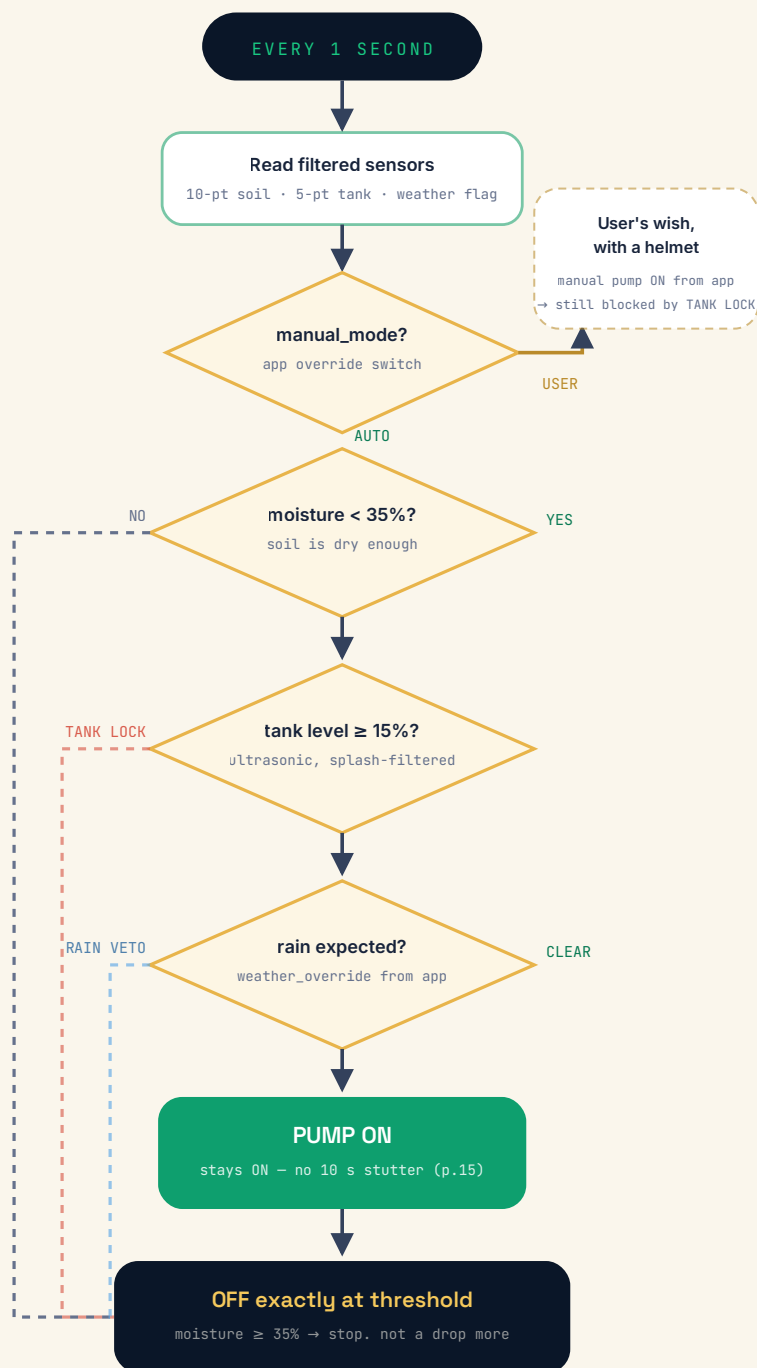
millis() NON-BLOCKING SCHEDULER

WATCHDOG: FED EVERY LOOP

LOGS: successful_uploads / failed_uploads

AUTO mode, drawn to scale

The whole watering brain in one diagram. Three locks must open before a single drop moves — soil says dry, tank says safe, sky says dry.



Three locks, one pump

- ◆ **Dry-soil consent** — watering starts only below the threshold you set (35% default).
- ◆ **Tank lock** — an empty tank means a dead pump if it ran dry. Level under 15%? Refused, even in



The light has its own brain

LDR says "dark" (past the 35 line)? The UV grow LED switches on — with **±2% hysteresis** so a passing cloud at dusk doesn't start a strobe party.

TESTED, NOT THEORISED

Pump AUTO ran a full **120-second soak** with zero glitches, then switched off **exactly** as moisture crossed the



```
// the whole law, in one line
pump_ON = (moisture <
moisture_threshold)
    && tank_safe
    && !weather_override;
```

LEGEND

● start / terminator

□ process

◇ decision

■ act (pump)

— refusal path → safe OFF

The bug that nearly sank us

AUTO mode was supposed to water until the soil hit 35%. Instead, the pump clicked on and off every ~10 seconds, like it had hiccups. This is the honest account of why.

THE SYMPTOM

tap. tap. tap.

In AUTO, the pump ran in ~10-second bursts instead of one continuous drink. The relay sounded like a telegraph key. Moisture climbed in zig-zags, not a slope.

THE FALSE LEADS

We blamed hardware. Of course we did.

Loose relay? Reseated. Power sag? Added the capacitor — it helped the rails, not the rhythm. Threshold noise? The filters were already clean. The stutter survived everything.

THE CLUE

The whole chip was rebooting, not the relay

We timestamped the serial log and saw it: a watchdog reboot every ~8–10 seconds, right in the pump's rhythm. The firmware wasn't deciding to stop — it was **dying mid-decision** and waking up in a different state.

THE ROOT CAUSE

17 blocking HTTPS calls per second

One Firebase call per sensor value, per control key, blocking, every loop. The network stalled, the loop starved, the 8-second watchdog fired, the chip rebooted, AUTO re-decided from scratch. On. Off. On. Off. **tap. tap. tap.**

THE CHAIN OF FAILURE



“

he watchdog wasn't the villain. It was the honest friend who slapped us awake every time the code froze.

— FIRMWARE LOG, THE NIGHT WE UNDERSTOOD

JSON bundling: the ~85% latency diet

Stop making one call per value. Pack everything into two envelopes — one up, one down. The fix took an evening; the lesson is permanent.

BEFORE • 17 CALLS / SECOND

17



- ◆ One blocking HTTPS call per metric & per key
- ◆ Network stalls starve the loop
- ◆ Watchdog reboots all day
- ◆ Pump: ~10-second stutter

AFTER • 2 CALLS / SECOND

2



- ◆ **1 PUT** — all 10 metrics to /sensors
- ◆ **1 GET** — all 9 keys from /controls
- ◆ Latency down ~85%, zero watchdog reboots
- ◆ Pump: ON until threshold, then off. Once.



```
// every second, exactly two envelopes:
```

```
PUT /sensors.json ← { moisture, temperature, humidity, light, tank_level,
                    lux, watchdog_status, voltage_sag, successful_uploads, failed_uploads }
```

```
GET /controls.json → { manual_mode, pump_state, light_manual_mode, grow_light_state,
                    capture_photo, moisture_threshold, tank_threshold, light_threshold, weather_override }
```

~85%

LATENCY CUT

0

WATCHDOG REBOOTS

120 s

PUMP SOAK • CLEAN

10+ min

TORTURE TEST • 0
CRASHES

09

A whisper goes up. A weather brain comes down.

Firebase is the single source of truth, the app is a single double-clickable file, and Delhi's sky gets a veto vote on every watering. Zero rupees, zero servers, zero excuses.

06a-div-cloud.html0

The whole truth, in one JSON tree

Every sensor reading, every command, every photo and log lives in one Realtime Database. The ESP32 writes, the app reads, and vice versa — they never talk to each other directly.



verde-tech-haha (Realtime Database • the single source of truth)

```

├─ sensors/      moisture • temperature • humidity • light • tank_level • lux
│                watchdog_status • voltage_sag • successful_uploads • failed_uploads
├─ controls/     manual_mode • pump_state • light_manual_mode • grow_light_state
│                capture_photo • weather_override
│                moisture_threshold • tank_threshold • light_threshold
├─ latest_scan/  imageUrl (base64) • status • captured_at
│                scientificName • diseaseName • probability • treatmentPlan
├─ weather/      city • temp • condition • description • humidity
│                wind_speed • rain_expected • synced_at
├─ historical_logs/ moisture_log [ { time, moisture }, ... ]
└─ actuators/    pump_actual • grow_light_actual • mode
  
```

RULES OF THE HOUSE

- ◆ Public read — a judge's phone works instantly
- ◆ Validated writes — booleans & numbers 0–100 only
- ◆ ESP32 authenticates with a legacy database secret

WHY REALTIME DB

Zero server code, free tier, JSON everywhere, and web sockets that make the dashboard feel alive instead of reloaded. The app and the ESP32 never talk directly — the tree is the conversation.

THE PHOTO PATH

JPEGs travel as base64 in /latest_scan via our Vercel upload API — no storage bucket, no extra bill. Photo to screen in under two seconds.

10

metrics
upstream every
second

9

command keys
downstream
every second

6

top-level
branches • zero
servers to
babysit



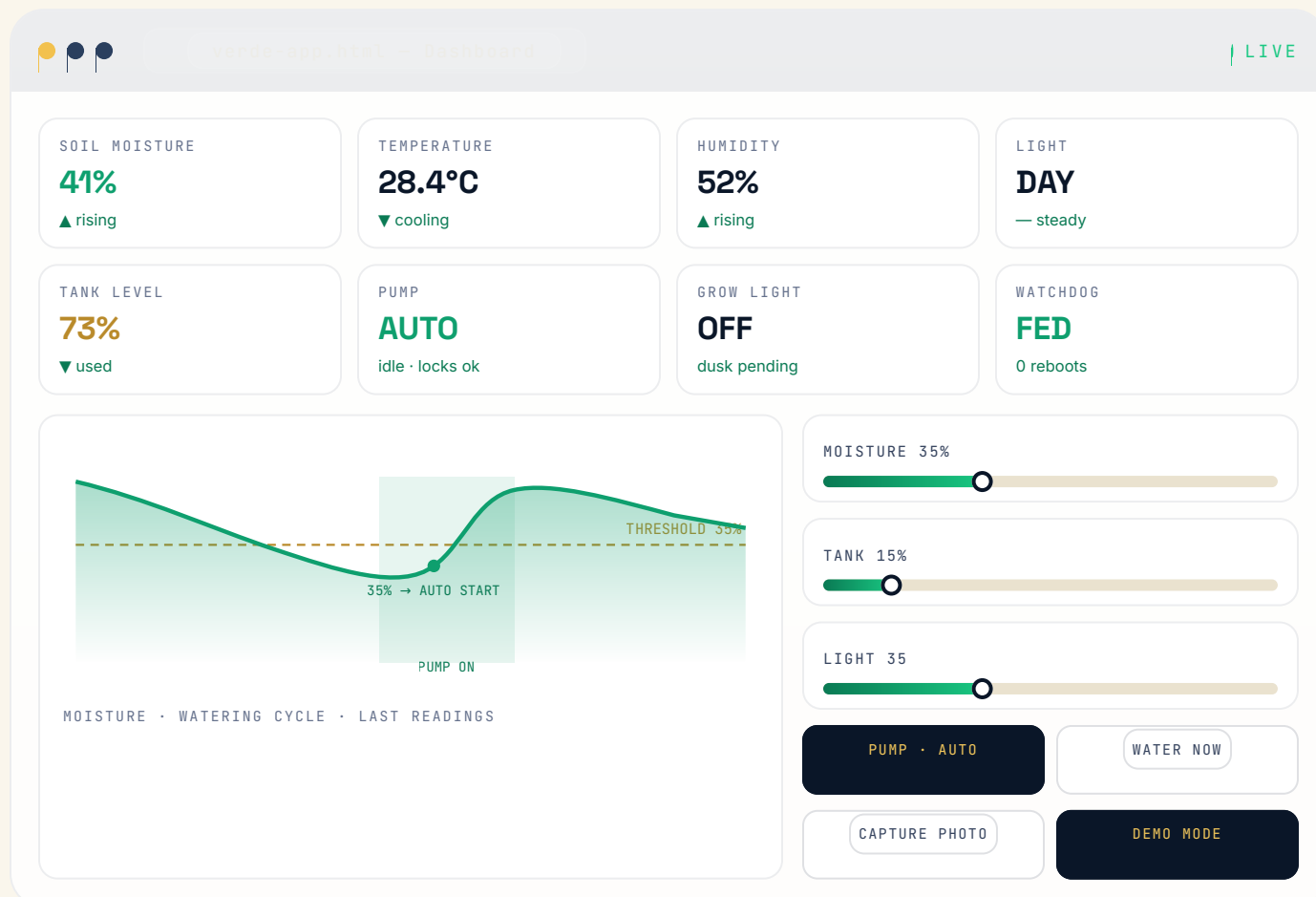
One tree, three readers

The ESP32 writes /sensors & reads /controls. The app does the reverse. The CAM only watches capture_photo. Nobody waits for anybody — they just check the tree.

One HTML file. Zero excuses.

SINGLE-FILE WEB APP

The entire product face — dashboard, weather, plant doctor, AI assistants — ships as one double-clickable HTML file. If a browser opens it, it runs. Recreated faithfully below.



Dashboard

8 live tiles with sparklines & hover trend arrows, 8 controls, 3 sliders, predicted actuator states, toasts, uptime, fullscreen demo mode.

Weather

Live Delhi sky, 5-day chips, rain-override with a visible countdown. Details next page.

Plant Doctor

CAM frame updating ≤ 2 s, CAPTURE button, upload-or-CAM modal, crop.health diagnosis + AI chat on the same photo.

AI Assistants

Gemini vision chat + OpenRouter sensor-aware chat with quick prompts and model fallback chains.

TANK CALIBRATION: SET EMPTY / SET FULL — APP-SIDE REMAP

IMAGE FLIP FIX (CAM MOUNTS UPSIDE-DOWN)

BURGER MENU • 4 PAGES • NO RELOADS

The sky gets a vote

Watering before a downpour is how roots drown. So the app asks Delhi's sky every three minutes — and can overrule the pump.

EVERY 3 MINUTES

The app looks up

An OpenWeatherMap call fetches Delhi's live condition + 5-day forecast: `GET /data/2.5/weather?q=Dehli`, key in the URL.

CONDITION DECODER

Rain codes translation

Weather IDs in the **2xx** / **3xx** / **5xx** / **6xx** families mean drizzle, rain, storm or snow. One flag becomes truth: `rain_expected`.

COUNTDOWN

The override arms visibly

A countdown timer shows on the Weather page. When it hits zero, the app writes `weather_override = 1` to the cloud.

THE VETO

Pump barred, politely

The ESP32's AUTO logic sees the flag and holds off — even with dry soil, the rain veto wins (page 14). When the sky clears, the veto retracts.



Verified live, not in theory

Real API, real city id 1273294, real 35 °C on the bench. The override fired on cue during testing and released the pump afterwards.



A commercial ₹8,000 kit would water on schedule straight into a monsoon. Verde's pump listens to the sky — **the forecast is a sensor too.**

CHECK CADENCE: 3 MIN

10

Point, click, and the plant gets a doctor.

crop.health reads the leaf, OpenWeatherMap reads the sky, and two assistants — Gemini and OpenRouter — refuse to stay dead thanks to thirteen fallback lifelines. The entire AI stack costs €0.

07a-div-ai.html0

Snap. Diagnose. Treat.

Point, click, and the plant gets a doctor who has seen millions of leaves. From shutter to treatment plan in about the time it takes to say "what's wrong with my monstera?"



T+0 S • CAPTURE

The eyes do their thing

App sets `capture_photo`. Within 1.5 s the CAM flashes, shoots SVGA JPEG and POSTs raw bytes to the Vercel upload API.

T+2 S • ON SCREEN

Base64 to the rescue

The photo lands in `/latest_scan`; the app frame refreshes — no storage bucket, no plugins, mounted-upside-down frame auto-flipped.

T+4 S • DIAGNOSIS

crop.health reads the leaf

POST `/api/v1/identification` with the base64 image returns crop + disease suggestions with probabilities and treatments.

THEN • CONSULT

The AI sees the same photo

REAL TEST RESULT • OUR TEST IMAGE

Nutrient deficiency — identified with a treatment plan

94%

Not a canned demo: our own photo went in, and `result.crop.suggestions[] / result.disease.suggestions[]` came back with name, **94% probability** and a step-by-step treatment plan that the app renders as a card.

AUTH: Api-Key HEADER

Four APIs, zero rupees

Each one researched, keyed, tested and given a fallback. Free tiers only — the entire software stack costs nothing to run.

API	WHAT VERDE USES IT FOR	AUTH	MECHANICS & ACCURACY NOTES
Open WeatherMap	Live weather + 5-day forecast → rain veto	KEY IN URL	GET /data/2.5/weather?q=Delhi — condition ids 2xx/3xx/5xx/6xx mean rain. Live-tested: Delhi 35 °C, correct city id 1273294.
crop.health (Plant.id)	Plant & disease identification from photos	Api-Key HEADER	POST /api/v1/identification with base64 image → crop.suggestions[] + disease.suggestions[] (name, probability, treatment). Named our test image's nutrient deficiency at 94% .
Google Gemini 2.5 Flash	Vision chat on the analysed photo	X-goog-api-key	POST /v1beta/models/gemini-flash-latest:generateContent with inline image + diagnosis + live telemetry. Lesson learned: "AQ" keys need the header; the pinned 2.5-flash id isn't offered to new users — the -latest alias is.
OpenRouter	Sensor-aware chat + vision fallback	BEARER sk-or-v1	OpenAI-compatible /chat/completions. 435 models reachable; free models rotate, so we built an 8-model text chain + 5-model vision chain — a 404 or 429 just promotes the next model.

₹0

MONTHLY RUNNING COST

Every tier free. Sustainability isn't a slide — it's the default.

4

APIS • 4 TESTED KEYS

Each key generated, authorised and exercised against the real endpoint.

13

MODELS IN FALLBACK CHAINS

If one model rotates off the free list, the twelfth backup still answers.

 **Why document the auth quirks?**

Because "key in URL vs header" cost us an afternoon each. The next team that builds on Verde should lose those afternoons to something more interesting.

Two assistants, thirteen lifelines

The app ships with two AI chats. One sees your plant, one reads its vitals — and behind both, fallback chains that refuse to dead-end.

**Gemini vision chat**

SEES THE PHOTO

Opens with the scanned image, its crop.health diagnosis and the live telemetry block, so answers ground themselves in your actual plant, your actual soil, your actual weather.

"tips are browning – is it the soil or the air?"
→ checks THIS photo • 94% diagnosis • 41% RH

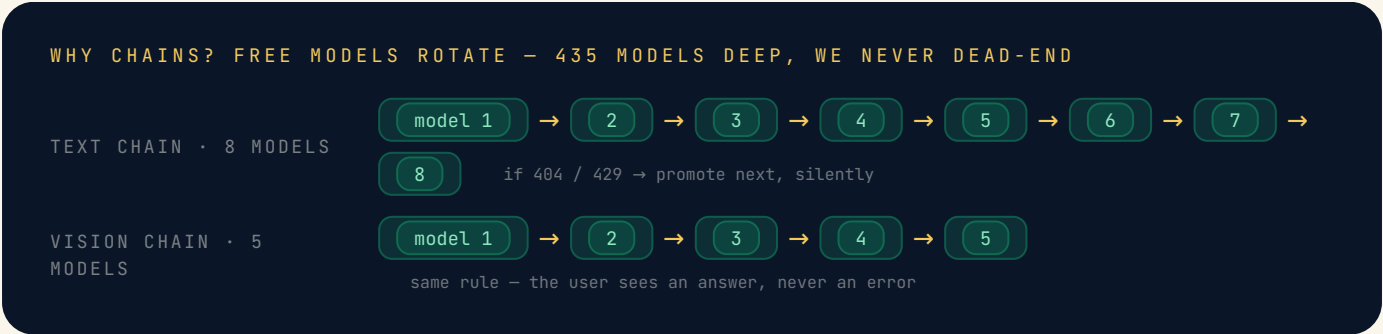
**OpenRouter sensor chat**

READS THE VITALS

A plant-care assistant wired into live sensor data with one-tap quick prompts: "Should I water today?", "Summarise this week's moisture," "Is my light schedule right?"

WATER TODAY? WEEK SUMMARY

LIGHT CHECK FEEDING PLAN



“Sorry, that model is unavailable” is a sentence our users will never read.

– DESIGN RULE #2

Everything on this wall **works.**

No roadmap items, nothing “coming soon”. Thirteen tests, thirteen passes, and a troubleshooting journal that counts all ten bugs out loud — because judges can smell perfect from a kilometre away.

08a-div-proof.html0

Everything on this wall works

No roadmap items, no "coming soon", no mockups. Each tile below was demonstrated live on the bench before it earned a place here.



Live dashboard

8 telemetry tiles, sparklines, hover trend arrows, uptime timer, fullscreen demo mode.



AUTO watering

Three locks — dry soil, safe tank, no rain — then waters to the exact threshold.



Rain veto

Delhi forecast checked every 3 minutes; an override countdown bars the pump.



Plant Doctor

Photo → crop.health → name, probability & treatment. 94% on our test scan.



On-demand camera

CAPTURE → SVGA photo in the app within 2 seconds, flash and all.



Vision AI chat

Gemini answers questions about the same photo the doctor just read.



Sensor-aware chat

OpenRouter assistant reads live vitals; 13-model fallback chains.



Live thresholds

Three sliders, adopted mid-loop by the ESP32, persisted in NVS flash.



Dusk grow light

UV LED rides the LDR with $\pm 2\%$ hysteresis. Zero flicker at sunset.



Tank calibration

SET EMPTY / SET FULL remaps the sensor in-app. No reflash, ever.



Watchdog + logs

8 s hardware fuse, upload success/fail counters, minute-by-minute history.



3-network fallback

Home, hotspot, school. It hunts for a signal so we never have to.

SENSORS 1 Hz

CLOUD 1 s

PHOTO $\leq$ 2 s

WEATHER 3 min

13/13 TESTS PASS

Thirteen tests. Thirteen passes.

✓ 13 / 13
PASS

We tested like the judges would try to break it — breath, hands, buckets and bad WiFi.

BOOT & STAMINA

WiFi / boot Joins the saved network cleanly from cold power.	✓ PASS
Watchdog soak — 10+ min Continuous uptime under load. Reboots: zero.	✓ PASS
Rain override Forced the flag; AUTO held even with parched soil.	✓ PASS

SENSORS

DHT11 breathe test One sigh on the sensor; humidity tile climbs, then settles.	✓ PASS
Moisture water-dunk Probe dunked; reading sweeps, filtered, no wild spikes.	✓ PASS
LDR cover test A thumb over the sensor = "dark"; grow light glides on.	✓ PASS
Ultrasonic hand test Hand moved in the tank; distance tracks, splash junk rejected.	✓ PASS

ACTUATION

Pump AUTO — 120 s soak Two minutes continuous watering. No glitch, no stutter.	✓ PASS
OFF exactly at threshold Moisture crossed 35% mid-run; pump stopped on the line.	✓ PASS
Tank lock Emptied the tank; both AUTO and manual were refused. Pump saved.	✓ PASS

EXPERIENCE

CAM capture ≤ 2 s Button to base64 photo on screen. Timed. Twice.	✓ PASS
Plant Doctor diagnosis Test scan identified at 94% with treatment plan.	✓ PASS
AI chats + fallbacks Both assistants answered; killing model 1 promoted model 2 silently.	✓ PASS



Method, for the record
Every test had a trigger and a pass condition written down before we ran it. "It worked once" is an anecdote; "13/13" is a result.

Ten bugs, zero hidden

Half our engineering education arrived wearing a bug costume. Published here for the next team — and for judges who like honesty more than perfection.

ELECTRICAL

02	Camera probe error 0x106	FPC ribbon unseated	Reseat gold-side down + full power cycle
03	"PSRAM not found" at boot	Weak power, not missing RAM	Dumb 5 V / 2 A adapter
04	Boot crash 0x20002	Camera + WiFi power surge at boot	Sequential boot: camera first, WiFi after 500 ms
05	WiFi dies when camera runs	20 MHz XCLK radio interference	Throttle XCLK to 8 MHz
06	67 W USB-PD brick starves board	PD needs a handshake chip we lack	Same 5 V / 2 A phone adapter
07	Relay dead on arrival... not	Split breadboard rails	Bridge rails: + to +, - to -
08	Temperature reads 0 °C	DHT on the wrong pin, floating ground	GPIO 4 + shared GND

SOFTWARE & CLOUD

01	AUTO pump clicks every ~10 s	17 blocking Firebase calls/s → watchdog loop	JSON bundling → 2 calls/s (page 16)
09	Firebase updates in "spurts"	13 calls/s of tiny writes	One bundled call per second
10	missing terminating quote compile error	Whitespace corrupted in a copy-paste	Re-download the source file; stop hand-copying



Pattern noticed: **six of ten bugs were power or connection**, not code. That's why page 11 exists — hardware discipline is firmware reliability.

ALL FIXED • ALL DOCUMENTED

12

It started as a
₹1,890 dare. It ends
where it grows **next.**

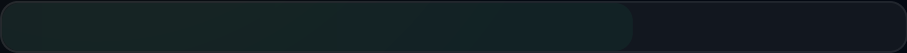
The full itemised audit against the ₹8,000 commercial box, why repairable-through-hole beats glued, and the seeds already ordered for the next build — solar, NPK probes, many plants.

09a-div-future.html0

₹1,890. Itemised, honestly.

Local-market prices, nothing hidden, software stack free forever. A product that runs for ₹0/month.

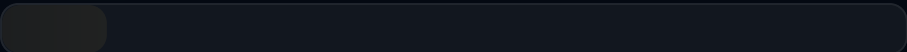
ELECTRONICS



₹1,320

ESP32, ESP32-CAM, five sensors, relay, pump, UV LED

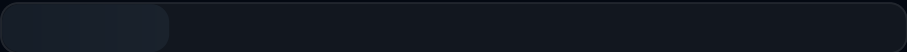
POWER & PROTECTION



₹220

5 V / 2 A adapter, capacitors, flyback diode

MECHANICAL



₹350

Breadboard, jumper wires, enclosure

SOFTWARE & APIS



₹0

Firebase, Vercel, OpenWeatherMap, crop.health, Gemini, OpenRouter — all free tiers

TOTAL BUILD

₹1,890

≈ \$23 USD

RUNNING COST

₹0 / month

VERSUS A ₹8,000 COMMERCIAL KIT — SAME JOB

VERDE

₹1,890

RETAIL KIT

₹8,000+

~76% CHEAPER · 4.2× VALUE

AND THE ₹8,000 KIT HAS NO CAMERA, NO AI DOCTOR

OURS OPENS, OURS TEACHES



Sustainability, for real

Repairable through-hole modules a student can replace in minutes. No subscription. No cloud bill. The most sustainable gadget is the one you can fix.

Where Verde grows next

Version 3 ships today. The seeds below are ordered by curiosity, not difficulty — several parts are already on our bench.

POWER

Solar autonomy

12 V panel + charge controller + battery. A plant that waters itself and powers itself.

SOIL

NPK probe

From "is it wet" to "is it fed" — nitrogen, phosphorus, potassium joining the metric bundle.

SCALE

Multi-plant zones

One brain, many pots: valves per zone, per-plant thresholds, a small farm on the balcony.

ALERTS

Telegram / WhatsApp

"Tank at 12%" straight to the family group chat — guilt-tripping, automated.

INTELLIGENCE

Predictive watering

Our minute-by-minute logs are training data: learn the drying curve, water before the wilt.

PLATFORM

Deployed Next.js app

The dashboard as a real hosted product — already scaffolded, waiting for a domain name.

“Every future item starts the same way version did: a sensor, a Saturday, and a bug to befriend.”

— THE ROADMAP, HONESTLY

The 3-minute stall tour

Exactly what we'd show you, timed to the second. Every beat is live hardware, live cloud, live AI — nothing pre-recorded.

- 0:00 • THE HOOK

"Plants die of silence. Watch one speak."

Breathe on the DHT11 — the humidity tile on the dashboard ticks up while you watch. The room is officially inside the system.
- 0:25 • THE DASHBOARD

Eight tiles, zero refreshes

Hover a sparkline for the last-10 trend arrow. Point out the predicted actuator states and the uptime timer still climbing.
- 0:50 • AUTO WATERING, LIVE

Drag the slider, hear the pump

Raise the moisture threshold above the current reading — AUTO consents, the relay clicks, water flows until the soil crosses 35%. Then it stops. Exactly there.
- 1:30 • THE RAIN VETO

The sky, overruled politely

Weather page: live Delhi forecast, the override countdown, and the moment weather_override would bar the pump despite dry soil.
- 2:00 • PLANT DOCTOR

Photo in 2 seconds, diagnosis in 4

CAPTURE fires the flash; the frame updates almost immediately; crop.health names the issue (94% in testing) with a treatment plan.
- 2:30 • THE AI CONSULT

Ask it anything about this plant

"Why are the leaf tips brown?" — Gemini answers from the same photo plus live telemetry. OpenRouter chat summarises the week if asked.
- 2:55 • THE CLOSE

The price reveal

"All of this — hardware, cloud, AI — cost ₹1,890 and runs for ₹0 a month. The ₹8,000 kit at the mall has no camera and no doctor."

If the venue WiFi dies

We switch to hotspot #2 (of 3 saved). If everything dies, AUTO keeps watering the plant offline and we demo the app on cached data — the cloud can leave, the watering can't stop.

PRE-DEMO CHECKLIST

TANK TOPPED

SOIL BONE-DRY (FOR THE DEMO)

HOTSPOT ON

APP FULLSCREEN

SPARE ADAPTER

It started as a ₹1,890 dare. It ended as a plant that talks.

Ten bugs befriended, one watchdog respected, four APIs tamed, thirteen tests passed — and a fern at home that hasn't been thirsty since March. Thank you for walking through it with us.



SCAN · DEMO AT THE
STALL

DESIGNED, BUILT, BROKEN & FIXED BY

Aarav Choudhary · Anuj

Class X · DAV ACON 5 — Tech Exhibition, 2026

FIRMWARE V3.0.7-FINAL

CAM V3.0.4-FINAL

13/13 TESTS PASS

₹1,890 · ≈ \$23

"The plant that waters itself — and talks to AI."